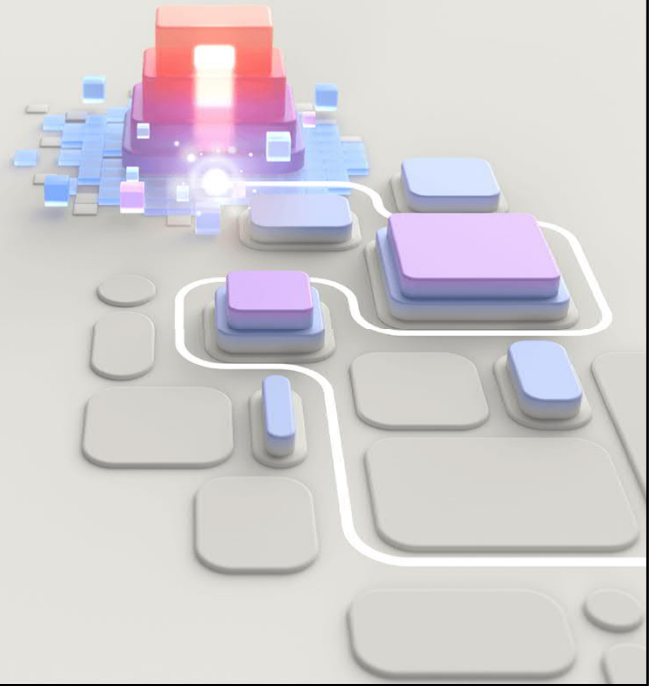




3. Management and governance

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<https://learn.microsoft.com/training/paths/describe-azure-management-governance>

Learning path 03—outline

You will learn the following concepts:

- 1 Cost management
 - Cost and pricing calculators
 - Cost management and tags
- 2 Governance and compliance
 - Blueprints, policies, and resource locks
 - Service Trust portal
- 3 Resource deployment tools
 - Portal, PowerShell, CLI, and others
 - Azure Arc and Azure Resource Manager
- 4 Monitoring tools
 - Azure Advisor, Azure Service Health, and Azure Monitor



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Cost management

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<https://learn.microsoft.com/training/modules/describe-cost-management-azure>

Cost management—objective domain

- Describe factors that can affect costs in Azure.
- Compare the pricing calculator and Total Cost of Ownership (TCO) calculator.
- Describe the Azure Cost Management Tool.
- Describe the purpose of tags.

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<https://learn.microsoft.com/training/modules/describe-cost-management-azure/1-introduction>

Factors affecting costs (part 1)



These are some of the factors affecting costs:

1) Resource type	2) Consumption	3) Maintenance
Costs are resource-specific, so the usage that a meter tracks and the number of meters associated with a resource, depend on the resource type.	With a pay-as-you-go model, consumption is one of the biggest drivers of costs.	Monitoring your Azure footprint and maintaining your environment can help you identify and mitigate costs that aren't necessary, such as shutting down underused virtual machines.

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<https://learn.microsoft.com/training/modules/describe-cost-management-azure/2-describe-factors-affect-costs-azure>

Factors affecting costs (part 2)



These are some of the factors affecting costs:

4) Geography	5) Network traffic	6) Subscription
The same resource type can cost different amounts depending on the geographic area, which has an impact on Azure costs.	While some inbound data transfers are free, the cost for outbound data or data between Azure resources is impacted by billing zones.	The type and configuration of your subscription can also impact your cost. For example, the free trial lets you explore some Azure resources for free.

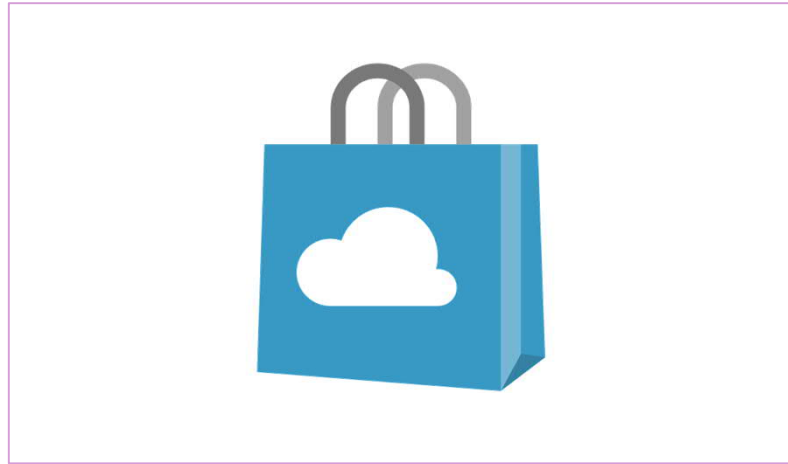
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<https://learn.microsoft.com/training/modules/describe-cost-management-azure/2-describe-factors-affect-costs-azure>

Explore Azure Marketplace

Azure Marketplace allows customers to find, try, purchase, and provision applications and services from hundreds of leading service providers, which are all certified to run on Azure.

- Open-source container platforms.
- Virtual machine and database images.
- Application build and deployment software.
- Developer tools.
- And much more, with 10,000-plus listings!



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<https://learn.microsoft.com/training/modules/describe-cost-management-azure/2-describe-factors-affect-costs-azure>

Connects end users with Microsoft partners, Independent Software Vendors (ISVs), and start-ups that offer solutions and services for Azure.

Azure customers, IT professionals and cloud developers can find, try, purchase, and provision Azure applications and services from certified service providers.

Includes close to 10,000 product listings.

There is also a Marketplace FAQ available at
<https://azure.microsoft.com/en-us/marketplace/faq/>

Azure Marketplace:

<https://azuremarketplace.microsoft.com/en-us/>

Pricing calculator

The pricing calculator is a tool that helps you estimate the cost of Azure products. The options that you can configure in the pricing calculator vary between products, but basic configuration options include:

- Region
- Tier
- Billing options
- Support options
- Programs and offers
- Azure dev/test pricing

The screenshot shows the 'Your Estimate' section of the Azure Pricing Calculator. At the top, it displays 'Virtual Machines' with a summary: '1 D2 v3 (2 vCPUs, 8 GB RAM) x 730 Hours'. It also shows 'Upfront: USD 0.00' and 'Monthly: USD'. Below this, there are several dropdown menus for configuration: 'REGION' (West US), 'OPERATING SYSTEM' (Windows), 'TYPE' ((OS Only)), and 'TIER' (Standard). Further down, there are 'CATEGORY' (All), 'INSTANCE SERIES' (All), and 'INSTANCE' (D2 v3: 2 vCPUs, 8 GB RAM, 50 GB Temporary storage, USD 0.209/hour). At the bottom, there is a quantity selector showing '1' and a duration selector showing '730' hours.

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<https://learn.microsoft.com/training/modules/describe-cost-management-azure/3-compare-pricing-total-cost-of-ownership-calculators>

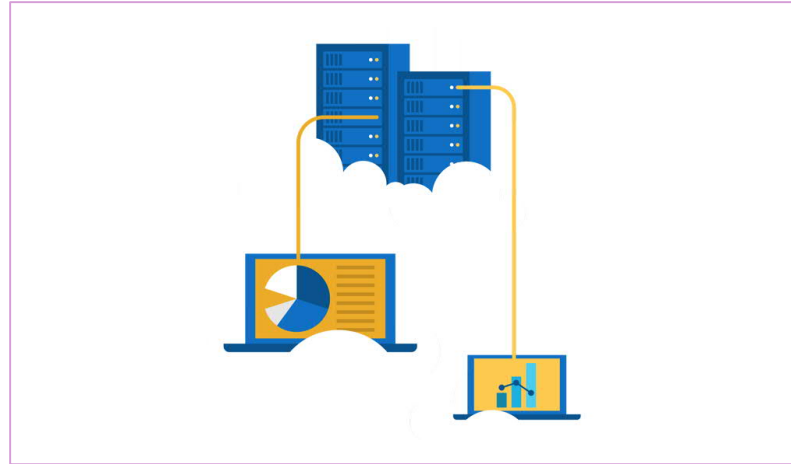
Pricing calculator: <https://azure.microsoft.com/en-us/pricing/calculator/>

Note: The pricing calculator provides estimates, and not actual price quotes. Actual prices can vary depending upon the date of purchase, the payment currency, and the type of Azure customer.

Exercise—use the Azure pricing calculator

Use the Azure pricing calculator to generate a cost estimate for an Azure virtual machine and related network resources.

1. Configure the pricing calculator.
2. Review the pricing estimate.

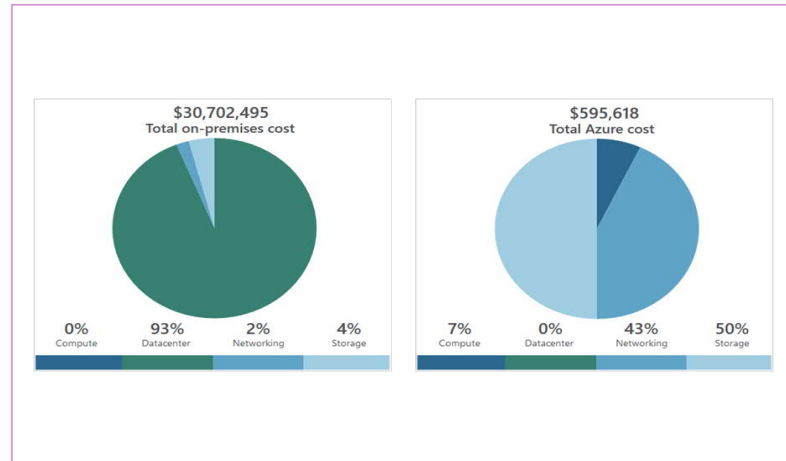


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<https://learn.microsoft.com/training/modules/describe-cost-management-azure/4-exercise-estimate-workload-costs-use-pricing-calculator>

Total Cost of Ownership (TCO) calculator

- A tool to estimate cost savings you can realize by migrating to Azure.
- A report compares the costs of on-premises infrastructures with the costs of using Azure products and services in the cloud.



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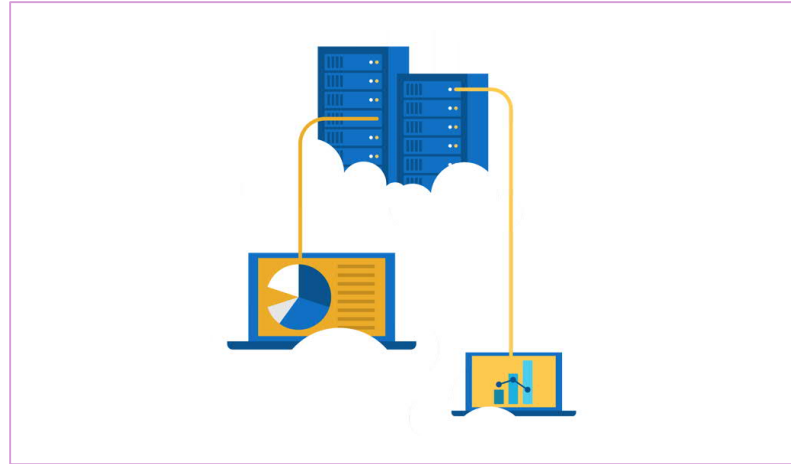
<https://learn.microsoft.com/training/modules/describe-cost-management-azure/3-compare-pricing-total-cost-of-ownership-calculators>

TCO calculator: <https://azure.microsoft.com/en-us/pricing/tco/calculator/>

Exercise—use the Azure TCO calculator

Use the Total Cost of Ownership (TCO) calculator to generate a cost comparison report for an on-premises environment.

1. Configure the TCO calculator.
2. Review the results and save a copy.

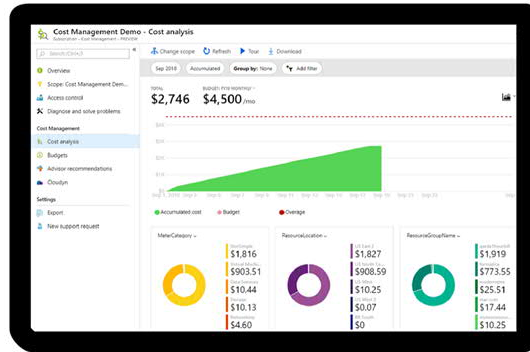


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<https://learn.microsoft.com/training/modules/describe-cost-management-azure/5-exercise-compare-workload-costs-use-total-cost-ownership-calculator>

Azure Cost Management

- Reporting: Billing reports
- Data enrichment
- Budgets: Set spend budget
- Alerting: When cost exceed limits
- Recommendation: Cost recommendations



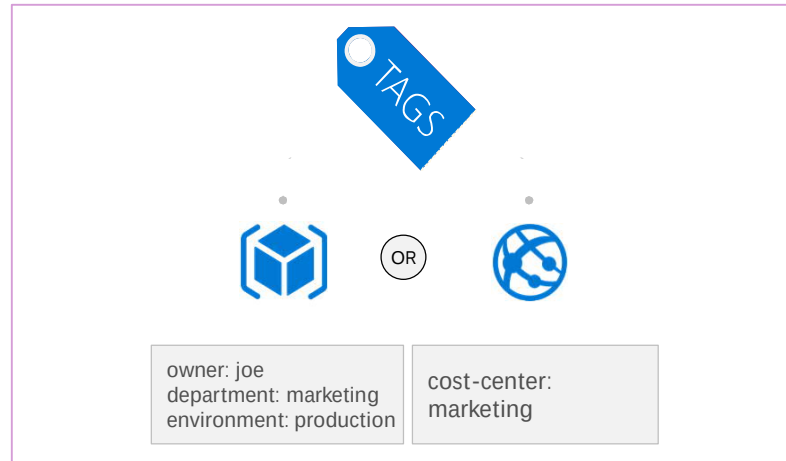
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<https://learn.microsoft.com/training/modules/describe-cost-management-azure/6-describe-azure-tool>

Note: For more information about Cost Management, go to <https://azure.microsoft.com/en-us/services/cost-management/>

Tags

- Provides metadata for your Azure resources.
- Logically organizes resources into a taxonomy.
- Consists of a name-value pair.
- Very useful for rolling up billing information.



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<https://learn.microsoft.com/training/modules/describe-cost-management-azure/7-describe-purpose-of-tags>

Tags: <https://learn.microsoft.com/azure/azure-resource-manager/management/tag-resources>

Governance and compliance

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<https://learn.microsoft.com/training/modules/describe-features-tools-azure-for-governance-compliance>

Governance and compliance—objective domain

- Describe the purpose of Azure Policy.
- Describe the purpose of resource locks.
- Describe the purpose of the Service Trust portal.
- Describe the purpose of Microsoft Purview.

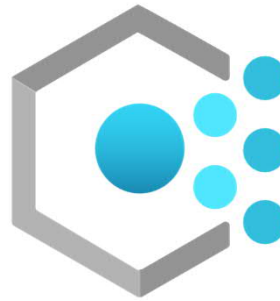
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<https://learn.microsoft.com/training/modules/describe-cost-management-azure/1-introduction>

Azure Policy

Azure Policy helps to enforce organizational standards and to assess compliance at scale. Provides governance and resource consistency with regulatory compliance, security, cost, and management.

- Evaluates and identifies Azure resources that do not comply with your policies.
- Provides built-in policy and initiative definitions, under categories such as Storage, Networking, Compute, Security Center, and Monitoring.



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<https://learn.microsoft.com/training/modules/describe-features-tools-azure-for-governance-compliance/3-describe-purpose-of-azure-policy>

Azure Policy: <https://azure.microsoft.com/en-us/services/azure-policy>

Stay compliant with your corporate standards and service level agreements (SLAs) by using policy definitions to enforce rules and effects for your Azure resources.

Evaluates and identifies Azure resources that don't comply with your policies.

Provides built-in policy and initiative definitions, under categories such as Storage, Networking, Compute, Security Center, and Monitoring.

Resource locks

- Protect your Azure resources from accidental deletion or modification.
- Manage locks at subscription, resource group, or individual resource levels within the Azure portal.

Lock Types	Read	Update	Delete
Delete	Yes	Yes	No
ReadOnly	Yes	No	No

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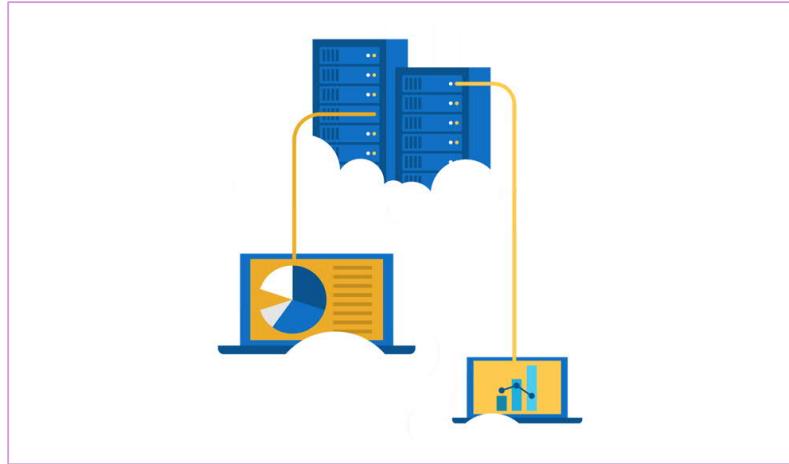
<https://learn.microsoft.com/training/modules/describe-features-tools-azure-for-governance-compliance/4-describe-purpose-of-resource-locks>

Resource Locks: <https://learn.microsoft.com/azure/azure-resource-manager/resource-group-lock-resources>

Walkthrough—manage resource locks

Create a resource and add a lock to prevent modification

1. Create a resource.
2. Add a ReadOnly resource lock to prevent resource modification.
3. Update lock and retest.
4. Remove the resource lock.
5. Delete the resource.



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<https://learn.microsoft.com/training/modules/describe-features-tools-azure-for-governance-compliance/5-exercise-configure-resource-lock>

NOTE: This is a bring-your-own subscription exercise. This will work fully in the Azure free trial, but the learner will need an actual Azure subscription for the exercise. May be better to do as a demo.

Microsoft Purview

Microsoft Purview is a family of data governance, risk, and compliance solutions that helps you get a single, unified view into your data. Microsoft Purview brings insights about your on-premises, multicloud, and software as a service data together.

- Automated data discovery
- Sensitive data classification
- End-to-end data lineage

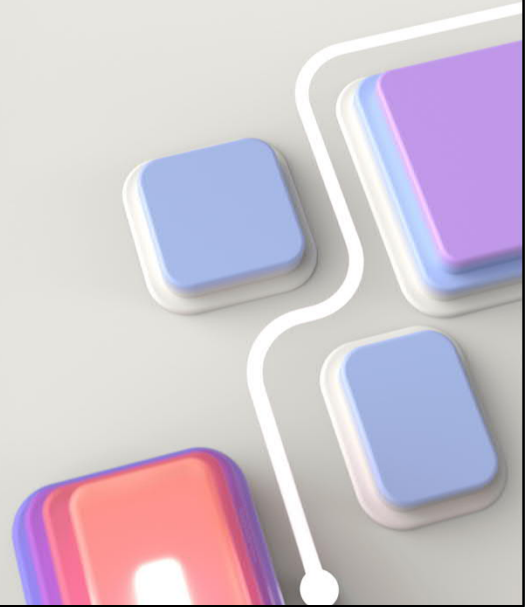


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Azure Policy: <https://learn.microsoft.com/en-us/purview/purview>

Management and deployment tools

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<https://learn.microsoft.com/training/modules/describe-features-tools-manage-deploy-azure-resources/>

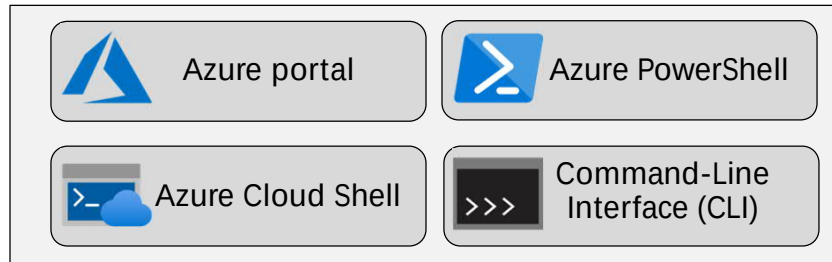
Management and deployment tools—objective domain

- Describe Azure portal.
- Describe Azure Cloud Shell, including Azure CLI and Azure PowerShell.
- Describe the purpose of Azure Arc.
- Describe Azure Resource Manager (ARM) and Azure ARM templates.

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<https://learn.microsoft.com/training/modules/describe-features-tools-manage-deploy-azure-resources/1-introduction>

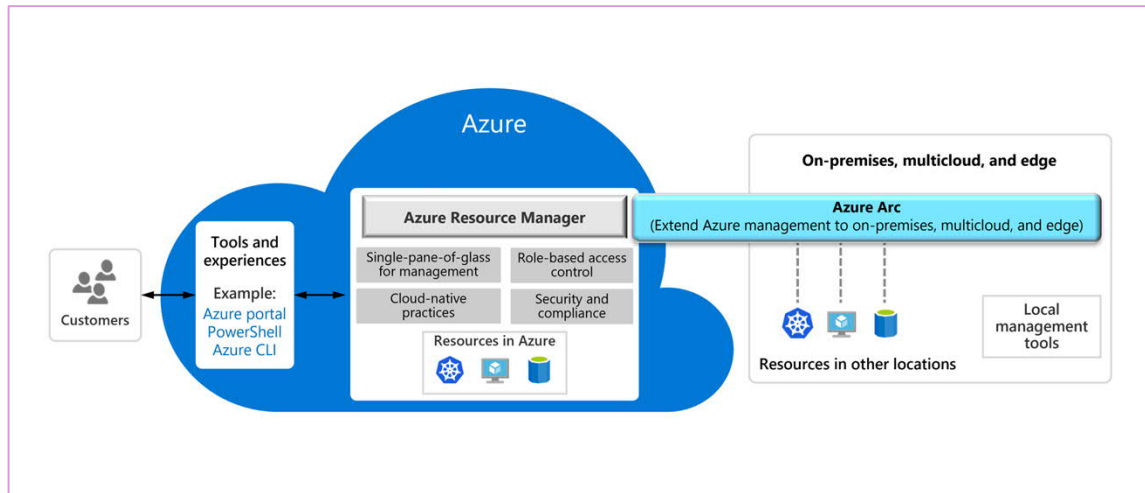
Tools for interacting with Azure



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<https://learn.microsoft.com/training/modules/describe-features-tools-manage-deploy-azure-resources/2-describe-interacting-azure>

Azure Arc

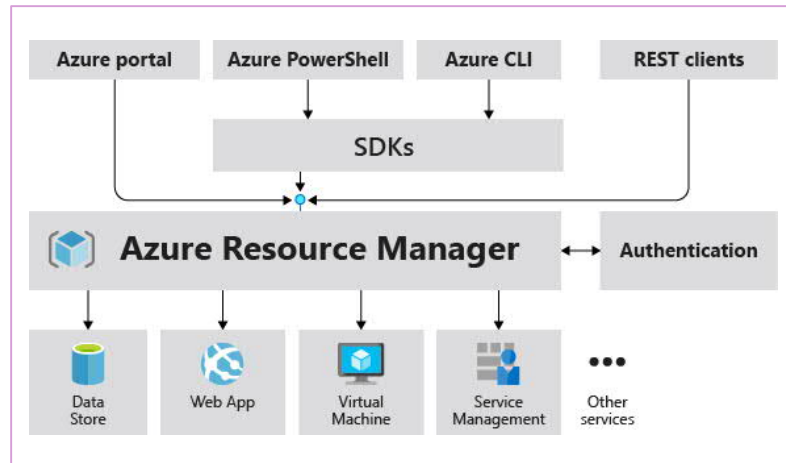


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<https://learn.microsoft.com/training/modules/describe-features-tools-manage-deploy-azure-resources/3-describe-purpose-of-azure-arc>

Azure Resource Manager

The Azure Resource Manager (ARM) provides a management layer that enables you to create, update, and delete resources in your Azure subscription.



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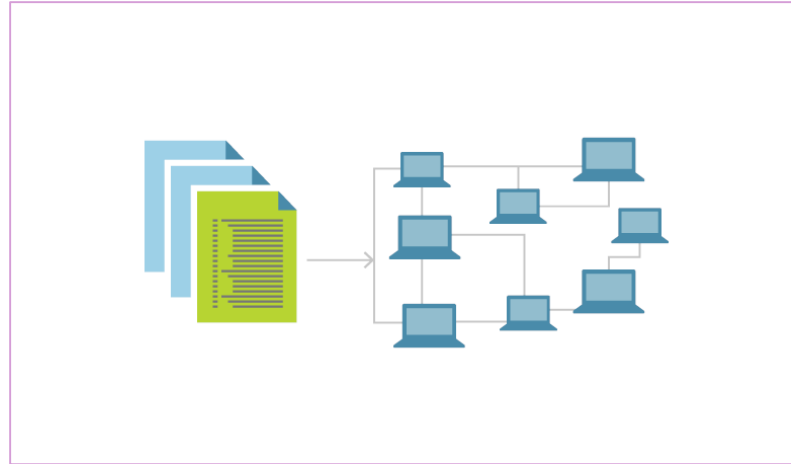
<https://learn.microsoft.com/training/modules/describe-features-tools-manage-deploy-azure-resources/4-describe-azure-resource-manager-azure-arm-templates>

- Provides a management layer that enables you to create, update, and delete resources in your Azure subscription.
- Create, configure, manage and delete resources and resource groups.
- Organizes resources.
- Controls access and resources.
- Automates using different tools and SDKs.
- Stores layouts in JSON files.

You can view more details about Azure Resource Manager at <https://learn.microsoft.com/azure/azure-resource-manager/>

Infrastructure as code

- Ensure consistency in deployment across your cloud ecosystem.
- Manage configuration at scale.
- Rapidly provision additional environments based on a standard configuration and build.



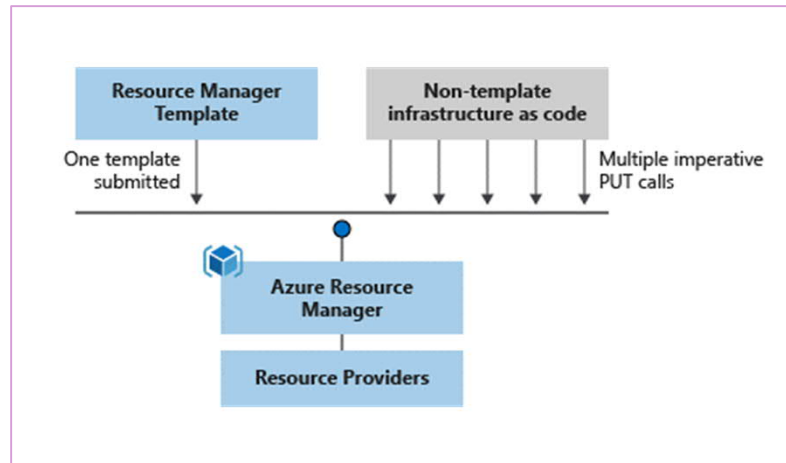
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<https://learn.microsoft.com/en-us/devops/deliver/what-is-infrastructure-as-code#deploy-iac-on-azure>

Azure Resource Manager (ARM) templates

Azure Resource Manager (ARM) templates are JavaScript Object Notation (JSON) files that can be used to create and deploy Azure infrastructure without having to write programming commands.

- Declarative syntax
- Repeatable results
- Orchestration
- Modular files
- Built-in validation
- Exportable code



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<https://learn.microsoft.com/training/modules/describe-features-tools-manage-deploy-azure-resources/4-describe-azure-resource-manager-azure-arm-templates>

<https://learn.microsoft.com/azure/azure-resource-manager/management/overview>
<https://learn.microsoft.com/azure/azure-resource-manager/templates/overview>

Deploy app resources Azure Resource Manager enables you to repeatedly deploy your app and have confidence your resources are deployed in a consistent state. You define the infrastructure and dependencies for your app in a single declarative template. This template is flexible enough to use for all of your environments such as test, staging or production. If you create a solution from the Azure Marketplace, the solution will automatically include a template that you can use for your app.

Organize resources

Azure Resource Manager makes it easy for you to manage and visualize resources in your app. You no longer have to deploy parts of your app separately and then manually stitch them together. You put resources with a common lifecycle into a resource group that can be deployed or deleted in a single action. You can see which resources are linked by a dependency. You can apply tags to resources to categorize them for management tasks, such as billing.

Control access to resources

With Azure Resource Manager, you can control who in your organization can perform actions on the resources. You manage permissions by defining roles and adding users or groups to the roles. For critical resources, you can apply an explicit lock that prevents users from deleting or modifying the resource. Azure Resource Manager logs all user actions so you can audit those actions. For each action, the audit log contains information about the user, time, events, and status.

Bicep

```
Bicep

param location string = resourceGroup().location
param storageAccountName string = 'toylaunch${uniqueString(resourceGroup().id)}'

resource storageAccount 'Microsoft.Storage/storageAccounts@2021-06-01' = {
  name: storageAccountName
  location: location
  sku: {
    name: 'Standard_LRS'
  }
  kind: 'StorageV2'
  properties: {
    accessTier: 'Hot'
  }
}
```

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<https://learn.microsoft.com/en-us/azure/azure-resource-manager/bicep/overview?tabs=bicep>

Azure monitoring tools

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<https://learn.microsoft.com/training/modules/describe-monitoring-tools-azure/>

Azure management tools—objective domain

Describe the functionality and usage

- Describe the purpose of Azure Advisor.
- Describe Azure Service Health.
- Describe Azure Monitor, including Azure Log Analytics, Azure Monitor Alerts, and Application Insights.

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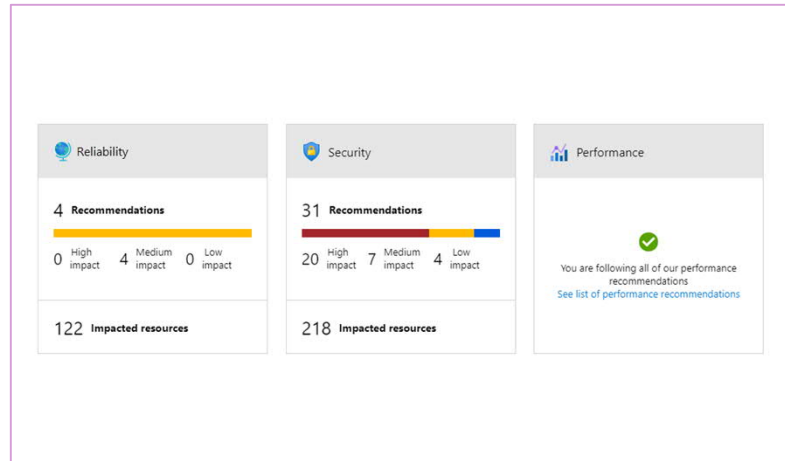
<https://learn.microsoft.com/training/modules/describe-monitoring-tools-azure/1-introduction>

Azure Advisor



Azure Advisor analyzes deployed Azure resources and makes recommendations based on best practices to optimize Azure deployments.

- Reliability
- Security
- Performance
- Cost
- Operational excellence



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<https://learn.microsoft.com/training/modules/describe-monitoring-tools-azure/2-describe-purpose-of-azure-advisor>

- Get proactive, actionable, and personalized best practice recommendations.
- Improve the performance, security, and availability of your resources.
- Identify opportunities to reduce your Azure costs.

Azure Advisor - <https://learn.microsoft.com/azure/advisor/advisor-overview>

Azure Service Health

Azure Service Health is a collection of services that keep you informed of general Azure status, service status that may impact you, and specific resource status that is affecting you.

Azure Status: Global view of the health of all Azure services across all Azure regions.

Service Health: Focused view on only the services and regions that you're using. If a service is experiencing a problem in a region you're not using, it won't show up here.

Resource Health: Tailored view of your actual Azure resources. It provides information about the health of your individual cloud resources.

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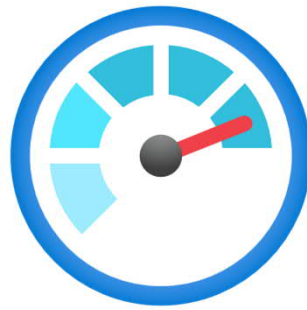


<https://learn.microsoft.com/training/modules/describe-monitoring-tools-azure/3-describe-azure-service-health>

Azure Monitor

Azure Monitor maximizes the availability and performance of applications and services by collecting, analyzing, and acting on telemetry from cloud and on-premises environments.

- Application Insights
- Log Analytics
- Smart alerts
- Automation actions
- Customized dashboards



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<https://learn.microsoft.com/training/modules/describe-monitoring-tools-azure/4-describe-azure-monitor>

Learning path 03 review



Microsoft Learn Modules
(learn.microsoft.com/training)

- Cost management
- Governance and compliance
- Resource deployment tools
- Monitoring tools

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<https://learn.microsoft.com/training/modules/describe-cost-management-azure/9-summary>

<https://learn.microsoft.com/training/modules/describe-features-tools-azure-for-governance-compliance/8-summary>

<https://learn.microsoft.com/training/modules/describe-features-tools-for-managing-deploying-azure-resources/6-summary>

<https://learn.microsoft.com/training/modules/describe-monitoring-tools-azure/6-summary>